SULAIMAN ALKADI

1200 E. California Blvd, Pasadena, CA 91125

EDUCATION

California Institute of Technology — Pasadena, CA

B.S. in Chemistry — Physical and Inorganic Concentrations — *Expected June 2027*

- Senior thesis: "Ultrafast Spin Dynamics in Tungsten and Molybdenum Complexes Relevant to Tungsto- and Molybdoenzymes" (Advisor: Prof. Ryan G. Hadt)
- Graduate coursework completed as an undergraduate: Quantum Mechanics, Molecular Spectroscopy, Advanced Ligand Field Theory, Advanced Inorganic Chemistry

HONORS & AWARDS

John A. Stauffer SURF Fellowship (twice; Gray and Hadt Laboratories)	2024, 2026
Ronan Armaan Mack SURF Fellowship (Nocera Laboratory, Harvard)	2025
KAUST Gifted Student Program (KGSP) Scholar; first Saudi student admitted to Caltech as an undergraduate	2023

RESEARCH EXPERIENCE

California Institute of Technology — Hadt Laboratory — Pasadena, CA

Undergraduate Researcher — *Sep 2024 – Present*

- Measured exchange coupling in distance-tuned dicopper(II) complexes by CW/pulsed EPR and spin-Hamiltonian modeling; found it lengthens low-temperature coherence by suppressing flip-flop dephasing.
- Designed and ran a six-month time-resolved Faraday rotation screen of 30+ compounds, identifying a series of optically addressable tungsten(V) spin qubits.
- Assigned the addressing transition to ligand-to-metal charge transfer by absorption, MCD and resonance Raman; demonstrated spin coherence in room-temperature solution with $T_2^* \approx 1$ ns.
- Built ORCA spin–vibronic models that traced room-temperature dephasing to five low-energy modes modulating the g-tensor; first author on the manuscript in preparation.

Harvard University — Nocera Laboratory — Cambridge, MA

Undergraduate Researcher — *Jun 2025 – Aug 2025*

- Showed with femtosecond and nanosecond transient absorption that dimerizing thioxanthone raises its triplet yield from ~70% to 98%, within two points of the theoretical maximum.
- Assigned an H-dimer from quenched fluorescence and a 47 → 690 ps singlet lifetime, and confirmed the exciton ordering with TD-DFT and exciton theory.
- Built a kinetic model showing that dimerization slows ground-state decay 200-fold but intersystem crossing only 10-fold, funneling excitations into the triplet manifold.
- Applied the dimer in photoredox catalysis to drive transformations that had resisted standard photosensitizers.

California Institute of Technology — Gray Laboratory — Pasadena, CA

Undergraduate Researcher — *Oct 2023 – Sep 2024*

- Tested whether Rehm–Weller back electron transfer reaches the Marcus inverted region, designing nanosecond transient-absorption experiments with pump-power and concentration controls against artifacts.
- Found rates plateau beyond the reorganization energy; computed DFT donor–acceptor couplings that place the reaction in the adiabatic regime, consistent with the missing inversion.
- Wrote a MATLAB kinetics platform: raw-file readers, log-time compression, and global multi-exponential fitting across wavelengths with confidence intervals on every rate constant.
- Summer 2024: synthesized a diiridium(I) photoreductant air-free and tested it as a one-/two-photon polymerization sensitizer; quenching showed it too weak for styrene or methyl methacrylate.

PUBLICATIONS

- Drummer, M. C.; Zhao, S.; Florescu-Ciobotaru, A.; Sarkar, A.; **AlKadi, S.**; Ghosh, R.; Nocera, D. G. Thioxanthone Dimers Exhibiting Near-Unity Triplet Quantum Yield and Thermally Activated Symmetry-Breaking Charge Separation. *Manuscript in preparation.*

PRESENTATIONS

- "All-Optical Spin Coherence in an Axial Tungsten(V) System of Relevance to the Active Sites of Tungstoenzymes," SURF talk, Caltech, Aug 2026.

- "Ultrafast Photophysics (and Photochemistry) of a Thioxanthone Dimer," SURF talk, Caltech, Oct 2025.
- "Development of One- and Two-Photon Radical Sensitizers," SURF talk, Caltech, Oct 2024.
- Two research group meeting presentations.

INDEPENDENT SOFTWARE PROJECTS

PyEasySpin — open-source Python port of the EasySpin EPR toolbox 2026
Python, NumPy/SciPy, Numba; github.com/Sulaimaninator/pyeasyspin

- Ported 300 EasySpin 6.0.12 functions to Python, matching MATLAB to one part in 10^6 on 25,000+ recorded calls.
- Parallelized spectral fitting across CPU cores; fits run 3.2–3.3× faster than EasySpin in MATLAB with identical results.
- Released on GitHub and PyPI (pip install pyeasyspin) so EPR simulation and fitting no longer require a MATLAB license.

SuperSpinner — spin-Hamiltonian parameters from CW-EPR spectra 2026
Python, PyTorch, MATLAB/EasySpin

- Trained a neural network that returns the g-tensor, metal hyperfine and linewidths, with calibrated uncertainties, from a raw CW-EPR spectrum in about 15 ms, replacing hours of starting-guess-dependent fitting.
- Generated 60,000 EasySpin training spectra after first checking which of its approximations hold: second-order perturbation theory is 15–23% wrong for Cu/V/Nb-sized hyperfine, so those cases are simulated by matrix diagonalization instead.
- Used as a starting point for EasySpin's esfit, it beat the textbook initial guess on all eight test spectra under the same time budget (8.5× lower g error); on real experimental data, refinement raised explained variance from 91% to 99.8%.

Starling — automated chirp correction for femtosecond transient absorption 2026
Python, NumPy/SciPy

- Replaced the hand-clicked chirp polynomial with a physical model: time zero is fit as the group delay through real optics (Sellmeier dispersion), with a standard error at every wavelength.
- Fit chirp, instrument response and kinetics jointly by variable projection, propagating chirp uncertainty into lifetimes.
- Reached a median time-zero error of 7.7 fs — 3.1% of the instrument-response width — across 40 randomized synthetic experiments, and flagged every run that exceeded the error threshold automatically.

TEACHING

California Institute of Technology Pasadena, CA
Teaching Assistant, Graduate Course — Ch 213a: Advanced Ligand Field Theory *Fall 2026*

- As an undergraduate, grade problem sets, hold office hours and lead discussion sections for this graduate-level course.

Caltech Y Pasadena, CA
Tutor *2024 – 2026*

- Tutored local middle- and high-school students in chemistry, math and physics for several hours each week.
- Developed lesson plans, practice problems and practice tests, and interactive ways to teach difficult concepts.

LEADERSHIP & SERVICE

Caltech Chemistry Club Pasadena, CA
President *Jun 2026 – Present*

- Lead a 30+ member club of undergraduates, graduate students, postdocs and Pasadena community members.
- Direct a K–12 STEM outreach program running demonstrations in regional schools and libraries.

Academic Director *Jun 2024 – Jun 2026*

- Helped organize the annual external-faculty lecture (since 2007; past speakers include Daniel Nocera, Chad Mirkin and Stephen Buchwald), bringing Paul Chirik (May 2025) and Robert Knowles (Feb 2026) of Princeton.
- Developed chemistry demonstrations and ran outreach at Arroyo Vista Elementary, Broadoaks School and Explore Caltech.
- Organized student–faculty events across the Chemistry and Chemical Engineering Division to widen student access to faculty.

Caltech International Student Programs Pasadena, CA
Orientation Leader & Student Advocate *2024 – 2026*

- Led international-student orientation for two years and advocated for international students year-round.

SKILLS

Languages Arabic (native), English (fluent)
Computational Python (SciPy, PyTorch, Numba), MATLAB, C++, R, Bash; DFT/TD-DFT in ORCA, Gaussian, Quantum ESPRESSO
Laboratory EPR, MCD, UV–vis, ultrafast TA, Faraday rotation; optical setups; air-free synthesis; protein expression